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Underground Mine Stability Assessment and GIS Mapping for Land Use Suitability.

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Presentation Layout

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Acknowledgements



Safety

Accident Free Days	167
LTI Free Days	289
FFPS	3164

Introduction

Geographic Information Systems (GIS)

are computer-based tools that capture, store, analyse, and display data related to positions on Earth's surface. Allow us to visualize, interpret, and understand spatial relationships, patterns, and trends by linking data to specific locations on Earth.

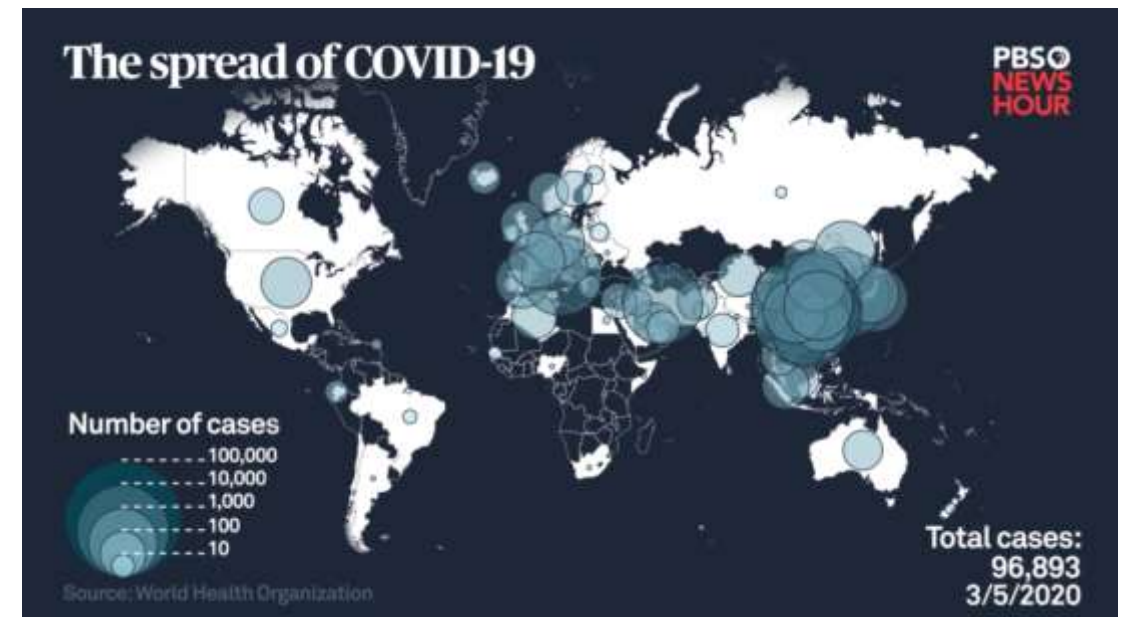
GIS = Mapping + Data + Analysis

Softwares which are used includes:

- ArcGIS
- QGIS
- GrassGIS

Real-Time Application

- Epidemiology
- Running tracking apps



Stability Assessment: Use GIS-based mapping to identify stable and unstable zones by analyzing the probability of stability in mined-out structures, predicting their integrity over the next 500 years.

Safety and Rehabilitation: Ensure safety for stakeholders and plan effective post-mining land rehabilitation.

Sustainable Development: Integrate real-time mining data for enhanced mine safety and promote sustainable land use.

Project aligns with Goedehoop closure objectives in a sense that the landscape for the community post mining needs to be left in a safe and stable conditions and also promote socio-economic opportunities.

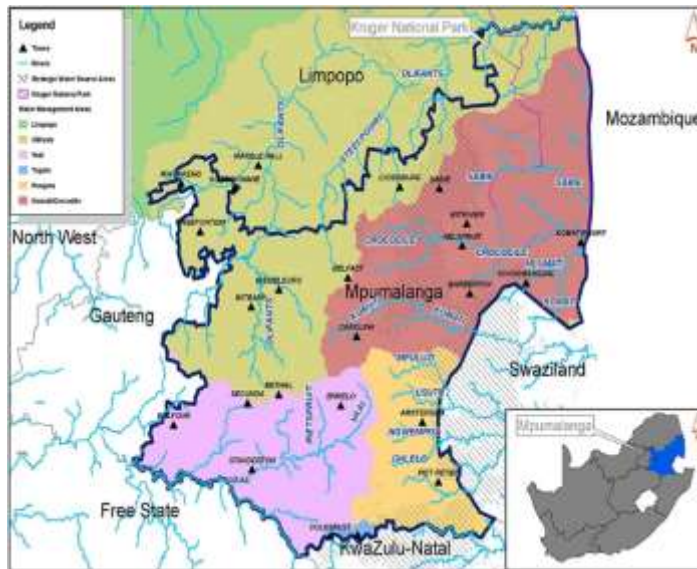
Project Background



Project Scope - Area

Geodehoop North Overview:

- Goedehoop consists of Goedehoop South and Goedehoop North, project is focused on the GH North.
- Goedehoop North operation is focused on extracting coal from 2 & 4 seam.
- The 5 seam workings were mined out in the 1950s and old 2 seam workings mined out in the 1970s , the lack of adherence to modern safety standards during historical mining activities has led to uncertainties regarding the stability of the underground workings.



Problem Statement



Previous mine stability assessments at Goedehoop North were based on predicted data and utilized a limited set of parameters. This approach may not accurately reflect the actual conditions within the mine. There is a critical need to incorporate actual mining parameters into stability assessments to ensure the safety of stakeholders and the integrity of mine structures post mining.

Parameters excluded:

- Mining centers.
- Mining heights.
- Bord widths.
- Mapped subsidence.



Parameters:

- Depth below surface.
- Competency of rock.

Tier 1 Hazard plan



Project Objectives



Primary Objectives:

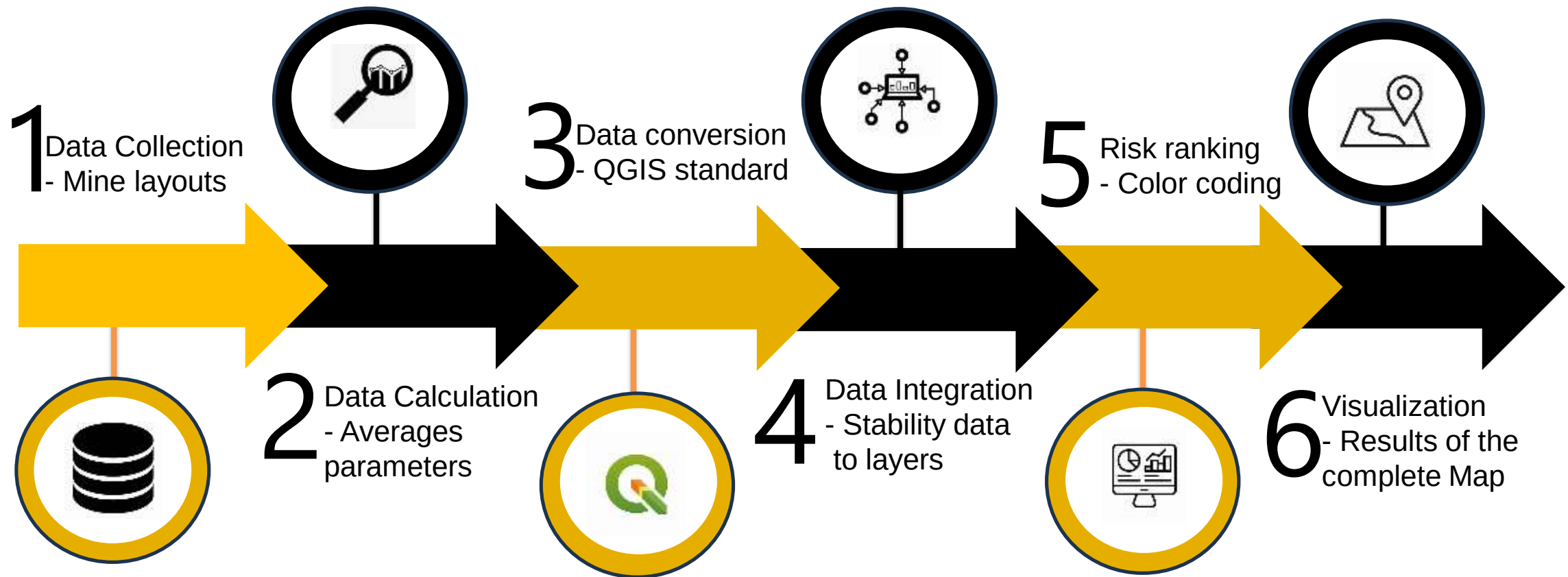
- **Underground stability analysis** to identify areas prone to subsidence and soil instability using risk-based modeling and spatial analysis techniques.
- **GIS-based Mapping** to identify and classify stable, moderately stable, and unstable zones in the mine.

Secondary Objectives:

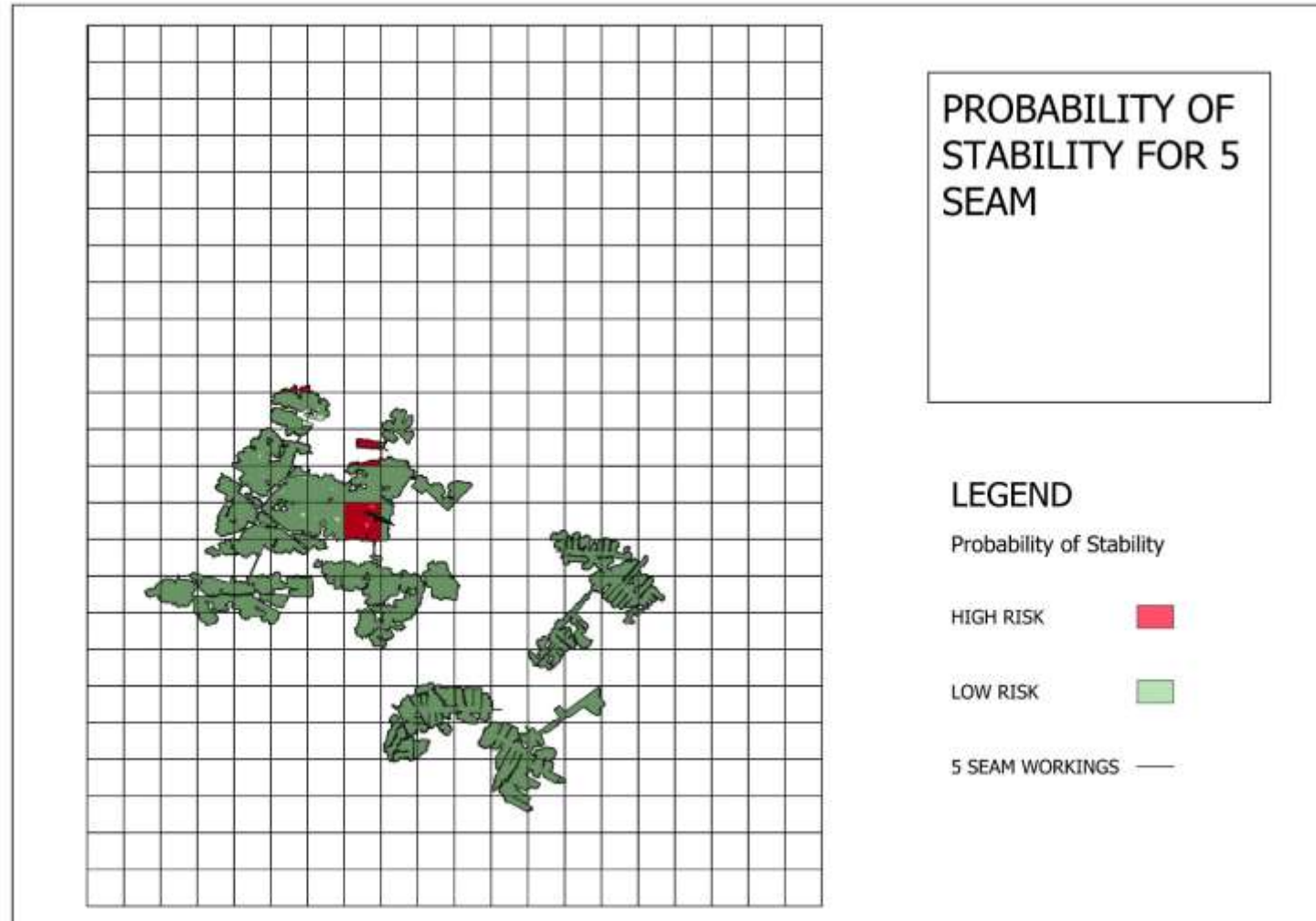
- **Evaluate land use suitability** of stable zones for post-mining development.
- **Propose rehabilitation practices** for unstable zones to ensure safe closure and future land use.



Methodology – Process Flow



5seam Stability Analysis

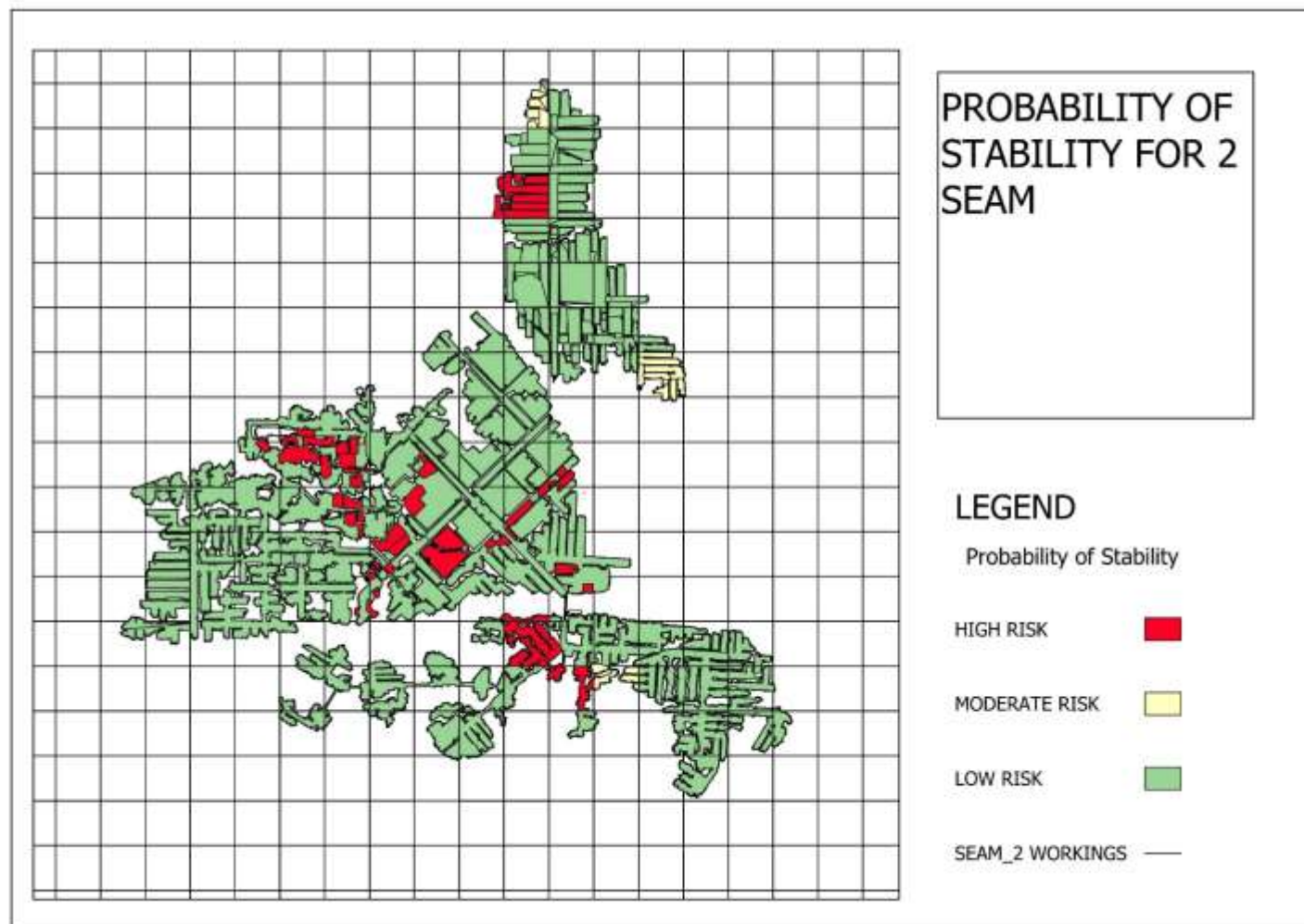


Total Area
1797ha

Low Risk
1719ha = 96%

High Risk
77ha = 4%

2seam Stability Analysis



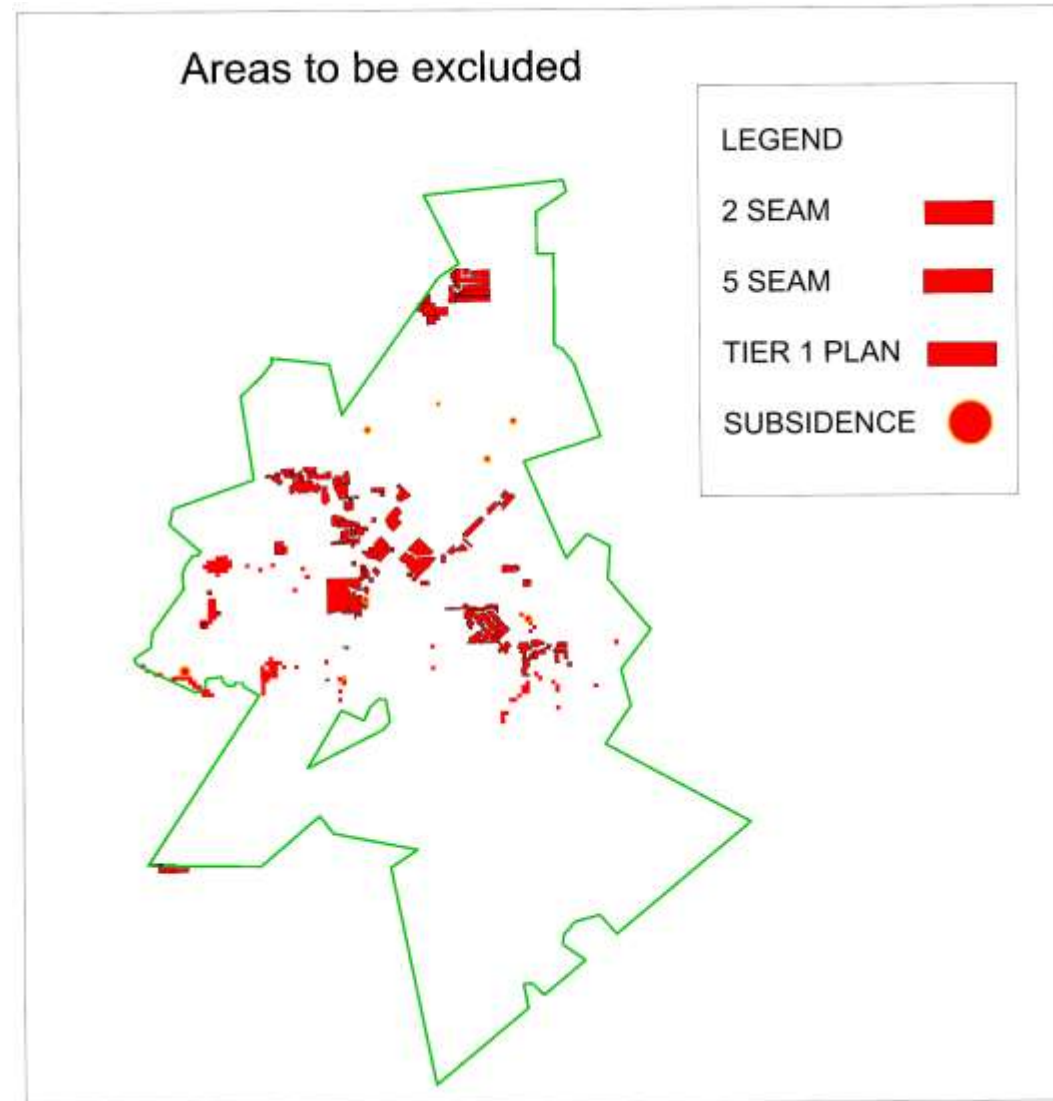
Total Area
3605ha

Low Risk
3222ha = 89%

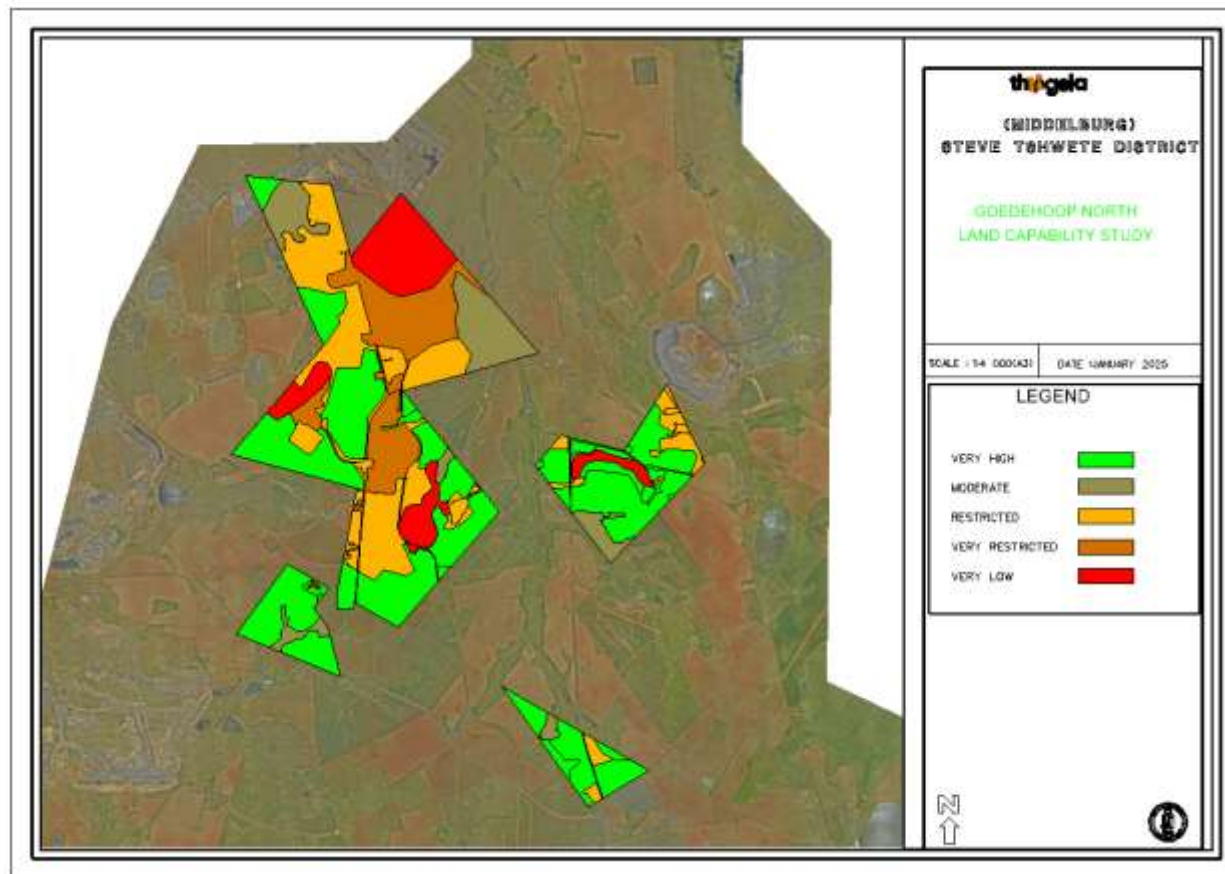
Moderate
22ha = 1%

High Risk
353ha = 10%

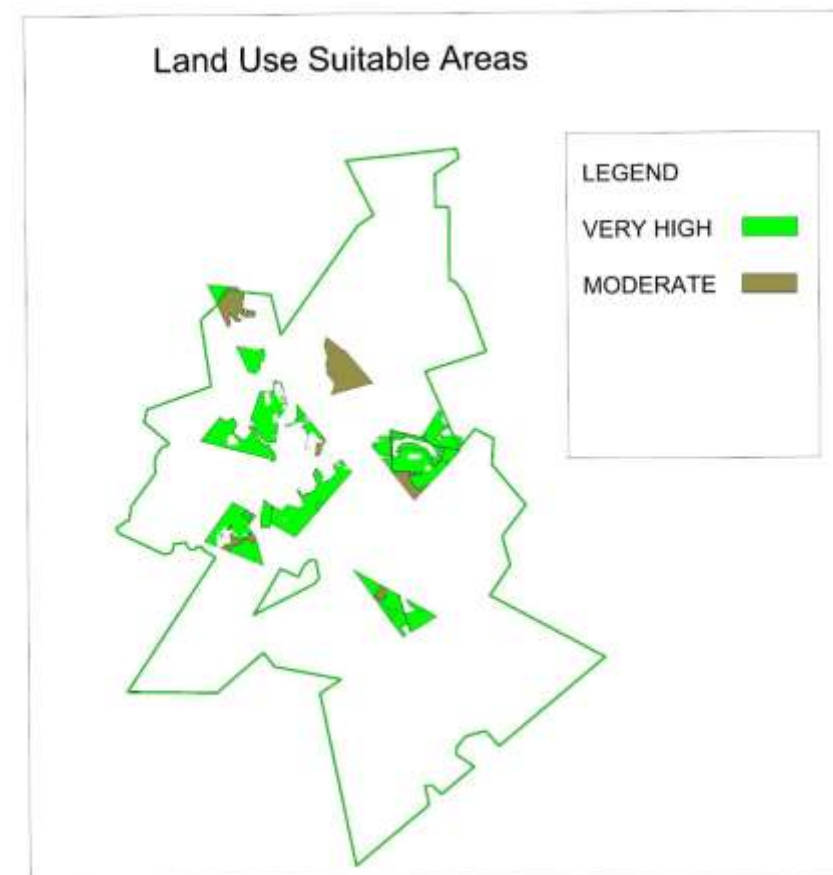
Unstable Areas to be excluded



Land Capability Study



Source: Goedehoop Colliery Soil, Land Use, and Land Capability Assessment Report



Treatment of unstable area

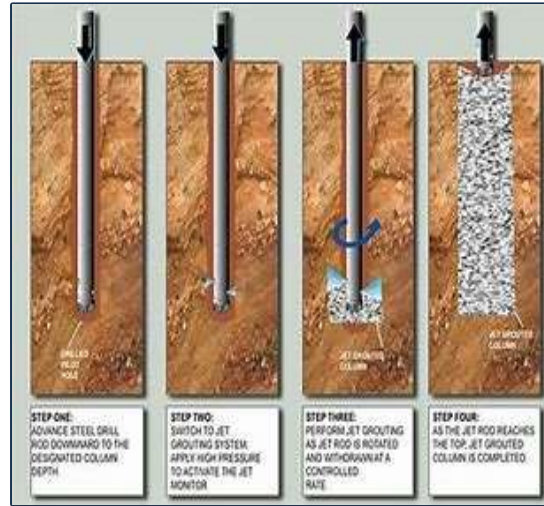


Current Mine Practices	Additional Mine Practices
Ground Movement Monitoring Systems	InSAR Monitoring System
Survey Note Notification	
Annual Lidar Survey	Semi-annual Lidar Survey
Drone Inspection (semi-annual)	Drone Inspections (quarterly)
Barricade	

Treatment of unstable area



Geosynthetic
Reinforcement for
Soil Stabilization



Surface Grouting
and Injection of
Stabilizing Agents



Revegetation and
Soil Reinforcement
with Native
Vegetation



Soil Compaction and
Reinforcement

Costs



Practices	Estimated Costs
Drone Inspections (quarterly)	R2 000 000
Semi-Annual Lidar Survey	R1 300 000
InSAR(Interferometric Synthetic Aperture Radar) Monitoring System	R2 000 000

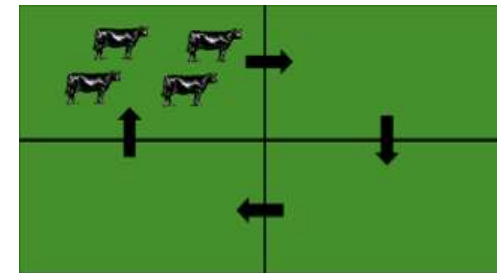
Total Budget (Current Mine Practices)	Total Budget (Additional Mine Practices)
R2 563 836	R5 300 000

Recommendations



To maximize the productive and safe use of land across various sectors, the following practices are recommended based on the stability of the land:

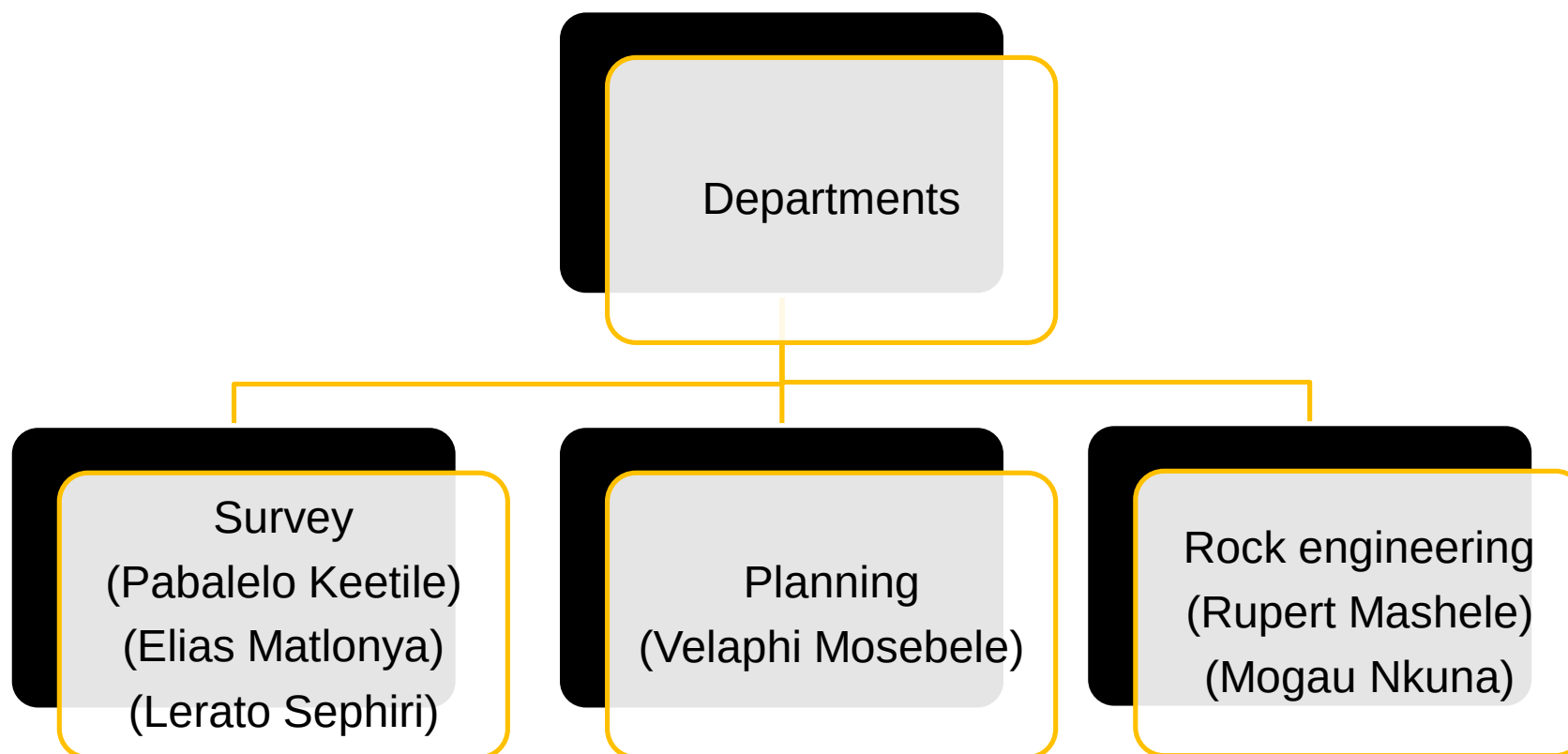
- **Stable Areas**
Diversified Land Use.
Ecological Integration.
Sustainable Construction.
- **Moderately Stable Areas**
Controlled Development.
Land Conservation.
Environmental Preservation.
- **Unstable Areas**
Strict Land Use Limitations.
Ecological Restoration.



Conclusion

- Areas prone to subsidence from the assessment were successfully identified enabling a clearer path to rehabilitation and post-mining land use.
- Currently, subsidence is not expected to pose a threat for up to 500 years to come, which is likely to occur in areas with low safety factors.
- This project serves as a model for integrating advanced geospatial technologies into mining rehabilitation efforts, proving that careful planning, strategic intervention, and innovative solutions can mitigate risks, promote safety, and contribute to sustainable development in post-mining landscapes.

Acknowledgements



Open for Questions.

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Thank you